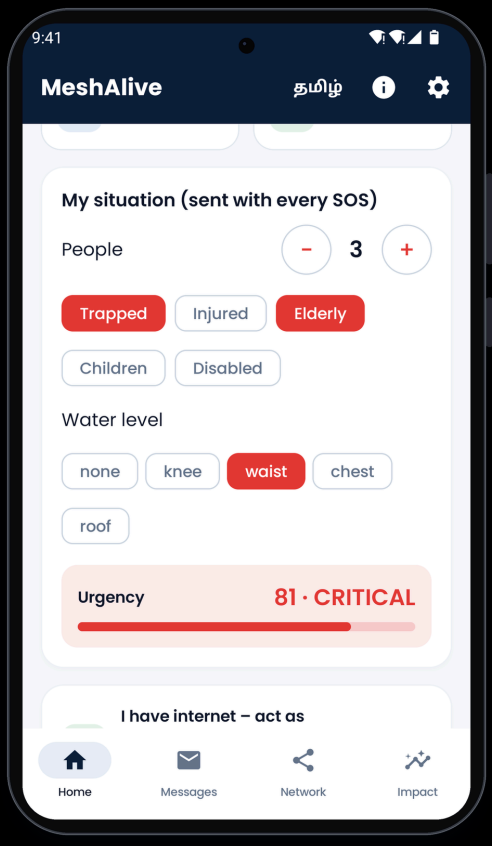
MeshAlive

CONCEPT DOCUMENT

MeshAlive

An offline disaster-response network that carries SOS messages to the district control room, and help back to people, when mobile towers fail.

Challenge	SevaFirst Innovation Challenge 2026
Theme	05 · Surakshit Bharat
Track	Track A · Product or Technology
Stage	Working prototype
Website and demo	meshalive.benzo.co.in
Source code	github.com/yogeshwardev/MESHALIVE-
Date	November 2026



Executive summary

Every cyclone and flood season, Indian districts lose mobile coverage for hours or days, at exactly the time when people most need help. People who are trapped cannot call for rescue, and district control rooms cannot see who needs help, where they are, or how urgent each case is.

MeshAlive keeps emergency communication working without towers, internet or new hardware. It is an Android app and a browser-based district dashboard. A person in danger sends a one-tap SOS that carries their GPS location and a short situation report, scored from 0 to 100 for urgency. Nearby phones relay the message over Bluetooth and Wi-Fi using battery-aware routing. When no route exists, rescue boats and teams carry the message out of the dead zone. The first phone that reaches signal uploads it to a live, urgency-sorted map in the control room. The control room's reply, such as "Rescue team on the way" or an evacuation alert, travels back through the same mesh and is digitally signed so it cannot be faked.

The prototype is complete and working: an Android app in Tamil and English, the control-room dashboard and 49 automated tests. In a 200-phone simulation, battery-aware routing delivered 100% of SOS messages with 147 times fewer radio transmissions than flooding. Citizens, rescuers and control rooms pay nothing for hardware. We are seeking a pilot partnership with a District Disaster Management Authority to field-test MeshAlive in one flood-prone ward.

100%	147×	₹0	49
SOS delivered in a 200-phone simulation	fewer radio transmissions than flooding	hardware cost for citizens and rescuers	automated tests passing

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1 The problem

Every cyclone and flood season, coastal and low-lying districts lose mobile coverage for hours or days. Towers lose power, backhaul links are cut, and the network that remains is overloaded. This happened in the 2015 Chennai floods and again during Cyclone Michaung in 2023.

1.1 Three gaps in today’s response

Gap	What happens today
People cannot call for help	Trapped families, the elderly, patients and people with disabilities cannot call, text or share their location.
Control rooms work blind	There is no view of who needs help, where they are, how many people are with them, or which case is most urgent. Boats and teams search by sight and word of mouth.
Existing apps are chat tools	Offline chat apps do not reach rescuers, do not prioritise urgent cases, drain the phones that relay for others, and cannot carry a reply back.

1.2 Who is affected, and where

People

- Families in flood- and cyclone-prone areas, especially the elderly, children, people with disabilities and patients
- Fishing communities and low-lying coastal settlements
- Rescue teams: NDRF, SDRF, fire and rescue services, boat operators, ambulance crews
- District Emergency Operations Centres and Disaster Management Authorities

Places

- Coastal districts of Tamil Nadu, such as Chennai, Cuddalore, Nagapattinam and Thoothukudi
- India’s cyclone-exposed east and west coasts
- River-flood regions: the Brahmaputra and Ganga basins, and Kerala
- Cities where flooding cuts power to mobile towers

1.3 Why it matters

In the first hours of a disaster, rescue capacity is scarce. Without location and urgency information, boats reach the people who are easiest to see rather than those in the most danger, and families who are already safe are searched for while others wait.

> PROBLEM STATEMENT
>
> When cyclones and floods take down mobile towers and power, people in danger cannot call for help, and district control rooms cannot see who needs rescue, where they are, or how urgent it is.

2 The solution

MeshAlive is an Android app and a browser-based district control room. Together they keep emergency communication working when the mobile network has failed, using only the phones people already own.

2.1 The rescue loop

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Ask for help	**Phones relay**	**Rescuers carry**	**Control room sees**	**Help returns**
One-tap SOS with GPS and a situation report, scored 0–100 for urgency.	Bluetooth and Wi-Fi mesh with battery-aware routing. Most urgent first.	Boats and NDRF teams carry stuck SOS out of areas with no signal.	The first phone with signal uploads to a live, urgency-sorted map.	Dispatch and evacuation alerts travel back to phones with no signal.

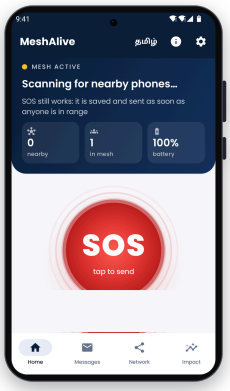
2.2 Key features

For people in danger

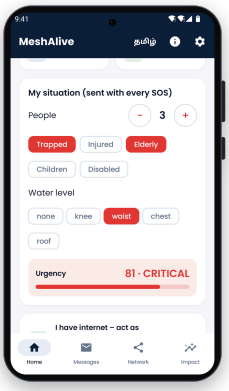
- 30-second guided setup: language, household and role
- SOS with a 3-second countdown and Cancel, to prevent false alarms
- Situation report: people, trapped, injured, elderly, children, disabled and water level, turned into an urgency score
- “I’m safe” check-ins, so rescuers can skip safe households
- “Rescue team on the way” banner, even with no signal
- Full SOS flow in Tamil and English

For rescuers and officials

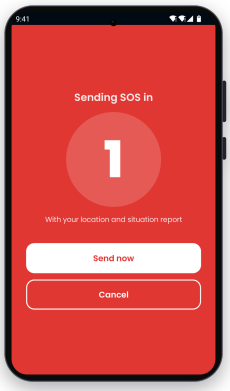
- Rescue-carrier mode for boats, NDRF/SDRF teams and ambulances
- Any phone that regains signal becomes a gateway automatically
- Live district dashboard: map, urgency, people count and route
- Dispatch team, mark rescued and evacuation broadcast, all digitally signed
- Field-trial evidence recorder and district coverage planner



Home: mesh status and one-tap SOS



Situation report: urgency 81, critical



3-second countdown with Cancel

Figure 1. Screens from the working Android app.

3 District control room

The dashboard runs in any browser on an existing control-room PC. Each SOS appears with its urgency, people count, situation details, GPS location and the route it took through the mesh.

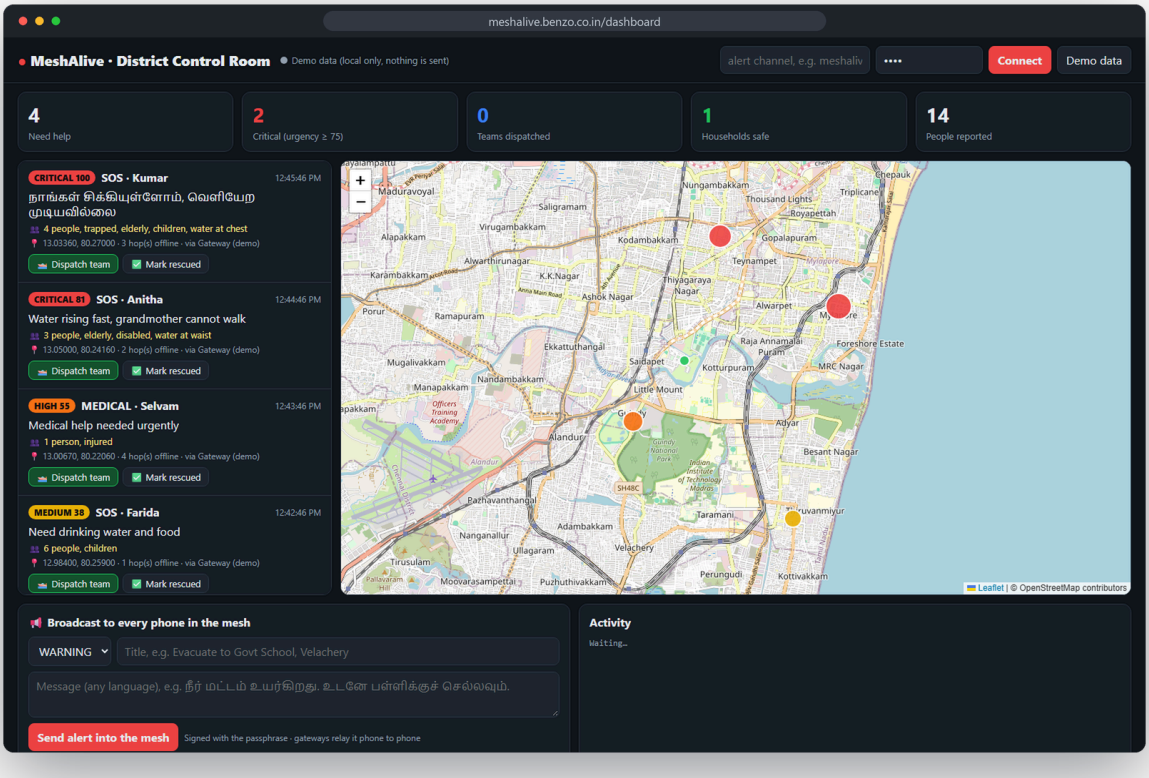


Figure 2. The district control-room dashboard, with SOS sorted by urgency on a live map of Chennai (sample data).

3.1 Operator actions

Action	What happens
Dispatch team	The operator types a note such as “Boat 7 on the way, stay on the roof”. A gateway pushes it into the offline mesh, the person’s phone shows **Rescue team on the way**, and the dashboard confirms that the phone received it.
Mark rescued	Closes the case on the map and tells the household that the rescue is recorded.
Evacuation broadcast	An alert in any language spreads phone to phone to every device in the mesh, and is shared again with phones that come into range later.
Signed commands	Every command is signed with HMAC-SHA256. Phones reject anything unsigned, altered or out of date, so a fake evacuation order cannot spread.

4 Innovation and comparison

Offline messaging already exists, but as chat between phones. MeshAlive is built as a complete rescue system, from the person in danger to the control room and back.

	Chat apps (Bridgefy, Briar)	LoRa radios (Meshtastic, goTenna)	MeshAlive
Reaches rescuers	No	No	**Live district dashboard**
Help back to the person	No	No	**Dispatch and evacuation alerts**
Prioritises urgent cases	No	No	**Urgency score 0–100**
No phone in range	Message waits	Message waits	**Rescue carriers carry it out**
Battery	Flooding drains phones	Flooding	**Battery-aware routing**
Extra hardware	None	₹2,500+ per person	**None**
Fake alerts	Not addressed	Not addressed	**Signed commands**
Language	English	English	**Tamil and English**

5 Technical approach

Layer	How it works
Discovery	Phones find each other over Bluetooth and Wi-Fi (Google Nearby Connections) and connect automatically, with no towers or internet.
Network map	Every few seconds each phone shares its battery, neighbours and internet status. Every phone rebuilds a live map of the mesh and measures link quality by round-trip time.
Routing	Battery-aware Dijkstra. Edge cost = link quality + battery penalty + hop cost. A virtual super-node joined to all gateways gives each phone its best next hop in one run, O((V + E) log V).
Priority and reliability	Urgency priority queue, store-and-forward retries, duplicate filter, hop limit (TTL) and delivery receipts with hop count.
Carriers and gateways	Custody transfer to rescue carriers. Any phone that gets internet uploads what it carries.
Security	AES-GCM encrypted messages; HMAC-SHA256 signed control-room commands.
Stack	Kotlin, Jetpack Compose, Room database, Google Nearby Connections; dashboard in plain HTML and JavaScript with Leaflet maps.

6 Prototype and evidence

The prototype is built and working. Routing was measured with the app's own engine on a simulated network of 200 phones sending 20 SOS messages.

Measure (same network for all methods)	Flooding	Shortest hop (BFS)	Battery-aware Dijkstra
SOS delivered	85%	100%	**100%**
Radio transmissions	14,533	88	**99**
Average hops	5.47	4.40	**4.95**
Battery left on the weakest phone	0.0%	3.2%	**6.3%**

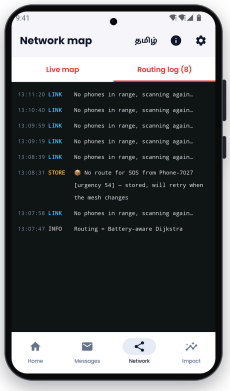
Battery-aware routing uses slightly more hops than shortest-hop routing, but leaves twice as much battery on the weakest phone, so the mesh stays alive longer.

6.1 What exists today

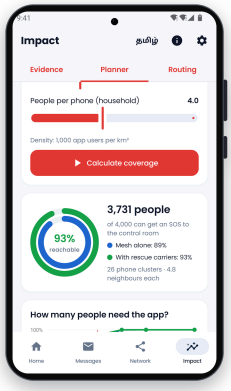
- Working Android app (APK) with all features described above
- Working control-room dashboard with signed dispatch and broadcast
- Public website with a live dashboard demo, and source code on GitHub
- 49 automated tests: routing, forwarding, duplicate filtering, TTL, triage, command signatures, coverage planner and evidence reports

6.2 Readiness for a field trial

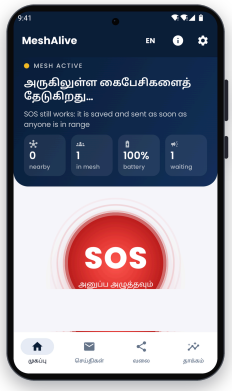
- Built-in recorder logs every real delivery: success rate, round-trip time, hops, phones met and relays
- One-tap export of a field-trial report for review
- Demo controls for in-room trials, such as cutting links and simulating low battery
- Next step: a structured pilot with 20–50 phones in one ward



Live routing log



District coverage planner



Home screen in Tamil

Figure 3. Evidence and planning tools inside the app.

7 Impact, cost and implementation

7.1 Beneficiaries and expected impact

- **People trapped or stranded:** their SOS reaches rescuers without towers, and they know help is coming.
- **Vulnerable groups:** prioritised automatically through the situation report.
- **Rescue teams:** exact locations, people count and urgency, instead of searching blind.
- **Control rooms:** a live, prioritised picture, and a channel to reach people without signal.

Coverage model. Using the app's own routing engine, a 1 km² ward with 1,000 app users (about 4 people per phone) lets around 3,700 of 4,000 people reach the control room. This is a model, to be validated in the pilot.

7.2 Cost and who pays

Stakeholder	Hardware	Notes
Citizens	₹0	Free app on their own phone. No SIM, data or account needed for the mesh.
Rescue carriers	₹0	Uses the phones that crews already carry.
Control room	₹0	Runs in any browser on an existing PC.
Running cost	Low	A small self-hosted alert server and app maintenance.
Rollout	Programme	Pre-monsoon awareness and installation drives, funded through the DDMA.

7.3 Implementation plan

Phase	Scope
1 · Field pilot 3 months	One flood-prone ward with the District Disaster Management Authority, local NDRF/SDRF boats as carriers and the dashboard in the district EOC. Measure delivery rate and SOS-to-dispatch time.
2 · District rollout	Installation drives through schools, colleges, panchayats, urban local bodies and self-help groups. The planner's adoption curve sets the install target for each ward.
3 · State and national	Distribution through the SDMA, dashboard feeds into state EOCs, alerts aligned with the national Common Alerting Protocol, and more Indian languages.

8 Adoption and support required

8.1 Who could adopt it

- District Disaster Management Authorities and District EOCs (primary users)
- State Disaster Management Authorities, such as TNSDMA, for state-wide rollout
- NDMA, for guidelines and integration with the Common Alerting Protocol
- NDRF, SDRF and fire and rescue services, as rescue carriers
- Urban local bodies and panchayats, for ward-level drives
- Ambulance services, fishing cooperatives and boat operators
- NGOs and volunteer networks such as Red Cross and civil defence; NSS and NCC units
- State IT departments and telecom operators, for hosting and integration

8.2 Support we are seeking

- A pilot partnership with a district administration and its DDMA, with access to the district EOC
- Collaboration with NDRF/SDRF and boat operators to field-test carrier mode
- Mentorship in disaster-management operations and public-sector deployment
- Technical guidance: security review, Common Alerting Protocol integration and self-hosting on government infrastructure
- Pilot funding for an alert server, test devices, and awareness and training drives
- Guidance on intellectual property, and incubation support

8.3 Links

Website and live demo	https://meshalive.benzo.co.in
Control-room dashboard	https://meshalive.benzo.co.in/dashboard/?demo=1
Android app (APK)	https://meshalive.benzo.co.in/download/MeshAlive.apk
Source code	https://github.com/yogeshwardev/MESHALIVE-

IN ONE LINE

MeshAlive turns the phones people already own into a rescue network, so that when the mobile network fails, help does not.